

COMMANDing anomalies: Continual video anomaly detection via dual-memory and temporal mamba modeling

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ABSTRACT

Weakly supervised video anomaly detection (WSVAD) aims to localize frame-level anomalies using only video-level labels, offering scalability for large-scale surveillance systems. However, existing methods often struggle to adapt to previously unseen and continuously evolving anomaly patterns, limiting their practical applicability. This challenge necessitates the development of continual learning (CL) frameworks that support incremental adaptation while preserving previously acquired knowledge. To this end, we propose a novel CL-based framework, dubbed **COMMAND**, for WSVAD that enables robust and adaptive anomaly detection in dynamic environments. **COMMAND** incorporates TempMamba, a temporal modeling unit based on Mamba blocks, which effectively captures both short-range and long-range temporal dependencies essential for distinguishing normal and abnormal behavior. In addition, MemDualNet introduces a dual-memory mechanism that retains both short-term variations and long-term contextual information, facilitating more expressive temporal representations. The framework Notation++, a continual learning strategy that integrates memory replay with a composite loss function comprising contrastive, focal, and multiple-instance objectives to alleviate catastrophic forgetting. Experimental results on benchmark datasets such as UCF-Crime and ShanghaiTech validate the effectiveness of the proposed approach, demonstrating superior performance in adaptability, generalization, and anomaly localization compared to existing state-of-the-art methods.

1. Introduction

The widespread adoption of video surveillance systems in public spaces has heightened the demand for automated anomaly detection methods capable of operating in real time [9,10,65,74,77–79]. Manual monitoring is impractical due to the sheer volume of video data generated across dynamic environments. While weakly supervised approaches [52] have advanced anomaly detection, they often fail to adapt to the continual emergence of novel and evolving anomalies in real-world scenarios. Despite recent progress, the challenge of learning incrementally from streaming, weakly labeled data remains understudied.

Video anomaly detection remains a fundamentally difficult task due to the infrequency of abnormal events and the substantial cost of acquiring frame-level annotations. Existing research predominantly

adopts two paradigms. Unsupervised approaches aim to learn representations of normal behavior from unlabeled data, identifying anomalies as deviations from this learned distribution [28,29,70,71]. Nevertheless, these methods often fail to detect subtle or context-dependent anomalies. This limitation is evident in scenarios such as slow fights or unauthorized loitering, where anomalous behavior may closely resemble normal activities, leading to misclassification [45]. Weakly supervised methods, in contrast, rely on video-level labels to reduce annotation costs while still training discriminative models [39,47,48,50,52,62]. However, they frequently encounter difficulties in accurately localizing the temporal boundaries of anomalies. For instance, when a lengthy surveillance video contains a brief anomalous segment, models may struggle to distinguish precise abnormal frames without fine-grained labels [15].

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Weakly supervised video anomaly detection methods commonly rely on general-purpose feature extractors to encode video segments and rank them according to anomaly scores [8,49]. Although these methods demonstrate strong performance on standard benchmarks, they encounter significant challenges in real-world surveillance scenarios. One primary limitation stems from the use of video-level annotations, which lack precise temporal information regarding the occurrence of anomalies. This imprecision can lead to incorrect interpretations, such as the misclassification of salient but contextually normal events as anomalous, especially in complex or crowded scenes. To mitigate this issue, Multiple Instance Learning (MIL) frameworks have been widely adopted [48]. MIL-based methods treat each video as a bag of segments, where only some instances may contain anomalous content, and optimize training by focusing on the most informative samples. Nevertheless, these approaches are prone to neglecting multiple co-occurring anomalies or emphasizing visually prominent yet semantically irrelevant segments. Such limitations hinder their effectiveness in handling the diversity and complexity of real-world video data.

In dynamic surveillance environments, the nature and distribution of anomalies are inherently nonstationary and evolve continuously over time. This poses a significant challenge for conventional models that are trained on fixed datasets, as their performance tends to degrade when exposed to novel or changing patterns of abnormal behavior. Continual learning [11] provides a principled solution by allowing models to incrementally learn from streaming data while retaining previously acquired knowledge. In real-world video anomaly detection systems, where new and unforeseen anomalies frequently emerge, continual learning is essential to maintain long-term model robustness and relevance [54,56]. However, applying continual learning in weakly supervised settings introduces several unique challenges. Unlike standard class-incremental scenarios, video anomaly detection relies primarily on the distinction between normal and abnormal events, with normal data serving as the essential reference for identifying deviations. Furthermore, the class imbalance and evolving nature of anomaly streams complicate the learning of stable and discriminative representations [5,51]. In addition, practical constraints such as data privacy and limited storage often make it difficult to retain or replay previous training samples [4,42]. Although progress has been made in integrating continual learning into unsupervised anomaly detection, its application in weakly supervised contexts remains relatively underexplored, particularly in scenarios involving emergent anomaly types. To address these challenges, we propose a continual learning framework for weakly supervised video anomaly detection using feature fusion, dual memory, Mamba-based temporal modeling, and replay-guided training for robust adaptability.

1.1. Motivation and contributions

The increasing deployment of large-scale video surveillance systems in public and critical infrastructure environments has emphasized the urgent need for automated anomaly detection techniques that are both scalable and adaptive. Traditional supervised methods typically depend on frame-level annotations, which are labor-intensive and impractical for long-term and wide-area monitoring. To alleviate this constraint, weakly supervised video anomaly detection techniques, such as the foundational work by [48], utilize video-level labels to reduce annotation overhead while maintaining competitive performance. In practical deployment, incoming video streams are accompanied by weak video-level labels, which provide sufficient guidance for continual learning while keeping annotation costs low. This makes long-term adaptation feasible without requiring detailed frame-level supervision. Further advancements, including RTFM by [52], have enhanced temporal modeling and feature discrimination. However, these approaches are commonly trained on static datasets and are not equipped to adapt to continuously changing environments where novel anomalies may

appear. This severely limits their deployment in real-world settings where anomalies are infrequent, varied, and unpredictable.

In practical surveillance scenarios, the nature of abnormal events is often unpredictable and continuously changing. Models trained in an offline manner struggle to generalize to novel situations, and they lack the flexibility to incrementally learn from new data while preserving previously acquired knowledge [24]. Furthermore, weak supervision introduces challenges in accurately identifying the temporal boundaries of anomalies, as video-level labels do not specify when the anomaly occurs. This can lead to incorrect identification of normal segments as anomalous. While continual learning [22,26] has shown promise in other fields for addressing such challenges, its integration into weakly supervised video anomaly detection remains largely unexplored. Motivated by these gaps, our work proposes a continual learning framework that combines augmented feature fusion, temporal modeling, memory retention, and replay-based training. Drawing inspiration from prior work such as [48,52], our approach is specifically designed to adapt to new anomaly patterns over time, ensuring robustness and long-term reliability in complex surveillance environments.

In a real-world deployment, video surveillance systems are not trained once and then left static. Instead, they operate for years, during which time new, previously unseen anomalies inevitably emerge. Conventional offline training strategies require collecting and re-annotating these new anomaly types, and then fully retraining the entire model from scratch. This process is computationally expensive, time-consuming, and can lead to significant periods of system downtime [55]. Continual learning, by contrast, provides a principled and cost-effective solution. By incrementally adapting to new data streams and retaining prior knowledge, a continual learning system can maintain high performance and relevance over long periods without the need for periodic and costly retraining [12].

To bridge this gap, this work introduces **COMMAND**, a novel **C**ontinual learning **f**ra**M**ework for weakly supervised video **a**no**M**aly **d**etect**I**on with TempMamba and MemDualNet, specifically designed to address the challenges posed by evolving and previously unseen anomalies in complex surveillance environments. Our method systematically integrates four key components, each addressing a critical limitation of current approaches and collectively advancing the state of the art in adaptability, robustness, and generalization. The core contributions are summarized as follows:

- We propose AugFuseNet, a multimodal feature fusion module that integrates spatial (RGB) and motion (optical flow) information. Through spatial augmentation strategies, AugFuseNet enhances the model's ability to generalize across varying environments, camera angles, and scene complexities.
- We propose TempMamba, a novel temporal modeling unit built upon Mamba blocks, designed to capture both short-range and long-range temporal dependencies. This enables more precise modeling of subtle and temporally extended anomaly patterns that are often missed by conventional recurrent or convolutional models.
- We design MemDualNet, a dual-memory representation module that stores both short-term and long-term contextual information. This architecture supports expressive, context-aware representations, allowing the model to recognize and distinguish normal versus abnormal behaviors in dynamically evolving scenarios.
- We introduce ContTrain++, a continual learning strategy that incorporates a memory replay mechanism along with a composite objective function. By combining contrastive learning, focal loss, and multiple instance learning, ContTrain++ enables incremental learning of new anomalies while effectively mitigating catastrophic forgetting.

The rest of the paper is organized as follows. [Section 2](#) reviews related work. [Section 3](#) presents the proposed framework. [Section 4](#) details the experimental results. [Section 5](#) concludes the paper.

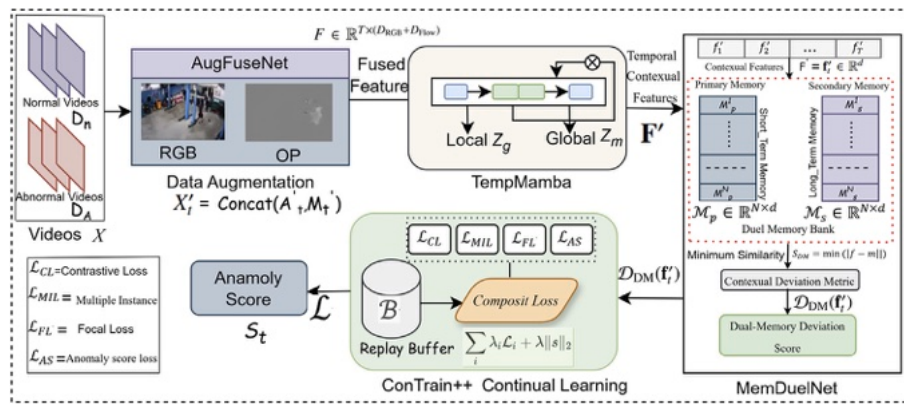


Fig. 1. Overview of the proposed method COMMAND. The figure shows a comprehensive workflow of the proposed model.

2. Related work

Weakly supervised learning has gained significant traction in anomaly detection due to its ability to train models using coarse labels (e.g., video-level annotations), substantially reducing manual labeling requirements. [48] pioneered this approach by developing a deep multiple instance learning (MIL) framework incorporating sparsity and temporal consistency constraints. Subsequent work by [72] employed graph convolutional networks to refine noisy supervision signals, enabling direct application of standard action classifiers without post-processing. The field advanced further with XD-Violence [60], which introduced both a large-scale dataset and a tri-branch architecture designed to capture diverse snippet relationships while supporting online detection. Addressing inherent limitations of weak supervision, [67] proposed innovative strategies including normalcy suppression mechanisms and clustering-based loss functions. Temporal modeling saw notable improvements through [76]’s integration of motion-aware features into augmented networks, complemented by [41]’s use of Siamese CNNs for patch-level anomaly detection through distance metric learning. The Robust Temporal Feature Magnitude (RTFM) method [52] enhanced MIL performance through dilated convolutions and self-attention mechanisms, while MIST [15] advanced self-training frameworks by combining pseudo-label generation with attention mechanisms. Spatial-temporal localization achieved greater precision through dual-branch relational reasoning networks [61] employing progressive refinement.

Recent multimodal approaches include VadCLIP [61], which leverages frozen CLIP models for dual-branch visual-language alignment, and TPWNG [63], which improves pseudo-label generation through CLIP-aligned text prompts with normality constraints. Video-based large language models [30] have further reduced annotation demands through long-term context modeling. Cross-domain generalization has been addressed by [23] through frameworks combining labeled source data with unlabeled external samples. Unsupervised methods continue to evolve, with MSFlow [75] introducing multi-scale flow modules for variable-sized anomalies and MambaAD [19] implementing Locality-Enhanced State Space modules for multi-scale dependency capture. GLAD [66] has advanced reconstruction quality through adaptive diffusion models trained with synthetic anomalies. Evaluation methodologies have matured considerably, with comprehensive assessments now incorporating advanced metrics, robustness analyses, and anomaly-type sensitivity [32]. The introduction of MVTec AD 2 [21] has addressed benchmark saturation through diverse industrial test cases. Domain-specific innovations include interpretable IIoT systems combining improved TCN-KAN VAEs with explainable AI [1], and FAUD [44]’s application of deep learning to thermal battery data for safety-critical monitoring.

Continual Learning (CL) enables anomaly detection models to adapt over time by incrementally learning from streaming data while preserving previously acquired knowledge. Moreover, in deep

reinforcement learning, CL strategies such as selective experience replay have been developed to mitigate catastrophic forgetting by retaining informative experiences across tasks [22]. Subsequently, replay-based methods like CLEAR [42] enhance continual learning by maintaining past knowledge and supporting new learning without task-specific information. Furthermore, Dark Experience Replay [4] combines rehearsal, knowledge distillation, and regularization to adapt to evolving data distributions even without explicit task boundaries. Interestingly, simple memory-based approaches such as GDumb [38] outperform more complex CL techniques, revealing limitations in current research. In addition, frozen vision-language models like CLIP [51] demonstrate strong zero-shot continual learning performance without fine-tuning. However, standard experience replay can cause overlap between new and old class representations, leading to forgetting; this has been addressed by asymmetric update rules that prioritize adapting new classes to existing ones [5]. Similarly, AttriCLIP [56] employs fixed encoders and learnable attribute prompts to classify new tasks without replay memory, outperforming prior methods. In the context of anomaly detection, lifelong learning is critical for continuous adaptation and retention despite challenges such as limited anomaly data and evolving distributions [13]. For example, CL-CaGAN [54] integrates continual learning to improve hyperspectral anomaly detection and cross-domain generalization.

Additionally, pixel-level anomaly detection methods adapted to continual learning settings have been validated on real-world datasets to establish robust benchmarks [3]. Recent works further advance this field: unsupervised continual anomaly detection frameworks such as UCAD [26] leverage contrastive prompting and structure-based learning to achieve superior detection without rehearsal. In network security, frameworks like TenaxDoS [2] combine federated and continual learning to enhance cooperative anomaly detection in Beyond 5G Open Radio Access Networks while preventing forgetting. Moreover, pyCLAD [14] offers a comprehensive software platform to design, evaluate, and benchmark continual anomaly detection strategies with reproducibility in mind. Finally, in malware classification, MADAR [40] addresses the unique diversity and replay challenges inherent to malware data, significantly outperforming previous continual learning approaches.

3. Proposed method

This paper introduces a novel continual learning framework, COMMAND, for weakly supervised video anomaly detection (WSVD), specifically designed to address the challenges of detecting previously unseen and evolving anomalies in real-world surveillance scenarios. As illustrated in Fig. 1, the proposed framework comprises four key components, each targeting a crucial aspect of the anomaly detection process: (1) AugFuseNet, a multimodal feature fusion module that combines RGB-based spatial appearance cues with optical flow-based motion information, enhanced through strategic data augmentation to improve

Table 1
Notation table.

Notation	Description
$F_i \in \mathbb{R}^{H \times W \times 2}$	Optical flow between (I_{t-1}, I_t)
$A_i \in \mathbb{R}^{d_a}$	Appearance features extracted from I_i
$M_i \in \mathbb{R}^{d_m}$	Motion features extracted from F_i
s_t	Clip at time t (segment of a video)
$\text{Flow}(s_t)$	Optical flow of clip s_t
x_t	Extracted feature for clip s_t
x'_t	Augmented feature
W_{in}	Input projection matrix
$Z = XW_{\text{in}}$	Projected feature sequence
Z_g, Z_m	Gated and modulated parts of Z
$Y_t = f(h_{t-1}, C_t)$	TempMamba output at time t
$F' = \text{CBAM}(\{y_t\})$	Attention-enhanced temporal features
f'	Feature after CBAM for scoring
M_p, M_s	Memory prototypes (primary and secondary)
$\{s_t\}$	Final anomaly scores for each segment
$\mathcal{L}_{\text{MIL}}$	Multiple Instance Learning loss
$\mathcal{L}_{\text{CL}}$	Continual learning loss
$\mathcal{L}_{\text{FL}}$	Feature-level contrastive loss
$\mathcal{L}_{\text{AS}}$	Anomaly scoring loss
λ_i	Coefficients to balance loss components

generalization across varying camera perspectives and environments; (2) TempMamba, a temporal modeling unit based on Mamba blocks, which captures both local and global temporal dependencies to better distinguish normal and abnormal temporal patterns; (3) MemDualNet, a memory-based representation module that employs a dual-memory structure to preserve both short-term and long-term contextual features, facilitating robust anomaly representation across dynamic scenes; and (4) ContTrain++, a continual learning strategy that integrates a replay mechanism with a composite loss function, consisting of contrastive, focal, and multiple instance learning objectives, to prevent catastrophic forgetting while ensuring adaptability to novel and evolving anomaly patterns. Collectively, these components form a unified and extensible framework that significantly advances weakly supervised anomaly detection in complex, real-world settings. Table 1 summarizes the key notations used throughout the methodology, while Algorithm 1 outlines the overall training procedure of the proposed framework.

3.1. AugFuseNet feature fusion module

To enable robust anomaly detection in surveillance videos under weak supervision, we propose AugFuseNet, a novel framework designed to generate discriminative spatiotemporal representations by integrating appearance and motion cues. Let I_t denote the RGB frame at time t . Appearance features $A_t \in \mathbb{R}^{d_a}$ and motion features $M_t \in \mathbb{R}^{d_m}$ are extracted from the RGB and optical flow modalities, respectively, using a pretrained I3D network [6]. Optical flow $F_t \in \mathbb{R}^{H \times W \times 2}$ is computed between consecutive frames (I_{t-1}, I_t) using the Farneback method to capture inter-frame motion dynamics. These appearance and motion features are concatenated to form a fused representation:

$$X_t = \text{Concat}(A_t, M_t) \in \mathbb{R}^{d_a + d_m}, \quad (1)$$

where X_t jointly encodes the spatial appearance (A_t) and temporal motion (M_t) information for the t -th video clip. This unified embedding preserves complementary multimodal cues, enabling the subsequent temporal modeling module (TempMamba) to effectively capture both short-term dynamics and long-term context essential for anomaly detection. To improve generalization across diverse camera perspectives and scene configurations, a data augmentation function $T(\cdot)$, including random spatial transformations such as rotations, is applied during training. The augmented inputs $T(I_t)$ and $T(F_t)$ yield transformed features A' and M' , which are similarly fused as $X' = \text{Concat}(A', M')$. This process enhances the model's ability to learn invariant multimodal representations and improves its robustness in real-world anomaly detection scenarios. Given a dataset of videos labeled as normal or anomalous, we define

Algorithm 1 Continual learning framework COMMAND for weakly supervised video anomaly detection.

Require: Dataset $D = \{x_i, y_i\}$; Replay buffer B ; Model parameters Θ ; Pretrained extractors $I3D_A, I3D_M$; Augmentation $T(\cdot)$

Ensure: Anomaly scores $\{s_t\}$

1: **Step 1: Feature Extraction and Augmentation**

2: **for** each $x_i \in D$ **do**

3: Segment into clips $\{s_t\}$

4: $x_t = [I3D_A(s_t, \text{RGB}); I3D_M(\text{Flow}(s_t))]$

5: $x'_t = T(x_t)$

6: **end for**

7: **Step 2: Temporal Modeling using TempMamba**

8: **for** each sequence $X = \{x_t\}$ **do**

9: $Z = XW_{\text{in}} = [Z_g, Z_m]$

10: $\tilde{Z} = Z_g \odot \sigma(Z_m)$

11: $Y_t = f(h_{t-1}, C_t)$

12: $F' = \text{CBAM}(\{y_t\})$

13: **end for**

14: **Step 3: Dual Memory Anomaly Scoring**

15: **for** each $f' \in F'$ **do**

16: $\text{score}_t = \min(\|f' - m^p\|, \|f' - m^s\|)$

17: **end for**

18: **Step 4: Continual Learning Optimization**

19: **for** each training step **do**

20: Sample current batch from D and replay batch B_r from B (selected by top anomaly scores, class-balanced, and temporally diverse)

21: Compute: $\mathcal{L}_{\text{CL}}, \mathcal{L}_{\text{MIL}}, \mathcal{L}_{\text{FL}}, \mathcal{L}_{\text{AS}}$

22: $\mathcal{L} = \mathcal{L}_{\text{MIL}} + \lambda_{\text{CL}}\mathcal{L}_{\text{CL}} + \lambda_{\text{FL}}\mathcal{L}_{\text{FL}} + \lambda_{\text{AS}}\mathcal{L}_{\text{AS}} + \lambda\|s\|^2$

23: Update Θ , memory banks, and buffer B

24: **end for**

25: **return** Anomaly scores $\{s_t\}$

two subsets $D_N = \{x_i\}_{i=1}^{N_N}$ and $D_A = \{x_i\}_{i=1}^{N_A}$, where each video x is segmented into T non-overlapping clips $\{s_t\}_{t=1}^T$, each comprising both RGB frames and optical flow data.

Features from each video segment are extracted using the pretrained I3D network, producing modality-specific feature tensors $F_{\text{RGB}} \in \mathbb{R}^{T \times D_{\text{RGB}}}$ for appearance and $F_{\text{Flow}} \in \mathbb{R}^{T \times D_{\text{Flow}}}$ for motion. These tensors are then concatenated along the feature dimension to form the combined feature matrix:

$$F = \text{Concat}(F_{\text{RGB}}, F_{\text{Flow}}) \in \mathbb{R}^{T \times (D_{\text{RGB}} + D_{\text{Flow}})}. \quad (2)$$

This joint representation F preserves both spatial and temporal information across all T clips, providing a unified embedding for the subsequent temporal modeling and memory-based modules in the framework. Random spatial augmentations may also be applied at the segment level during training to further improve generalization. The resulting subsets D_N and D_A are then used to train the anomaly detection model under weak supervision, where only video-level labels indicating normality ($y = 0$) or anomaly ($y = 1$) are available, thus removing the need for detailed frame-level annotations.

3.2. TempMamba temporal modeling unit

To model temporal dependencies within the fused spatiotemporal feature sequence, we propose the TempMamba module. We employ a Mamba-based architecture to overcome the limitations of conventional methods: unlike RNNs and LSTMs that struggle with long-range dependencies, and Transformers that are computationally expensive for long video sequences, Mamba's efficient Selective State-Space Model allows it to effectively capture both short- and long-term event dynamics. The module processes the input through a series of Mamba blocks and refines the features using a Convolutional Block Attention Module (CBAM)[58]

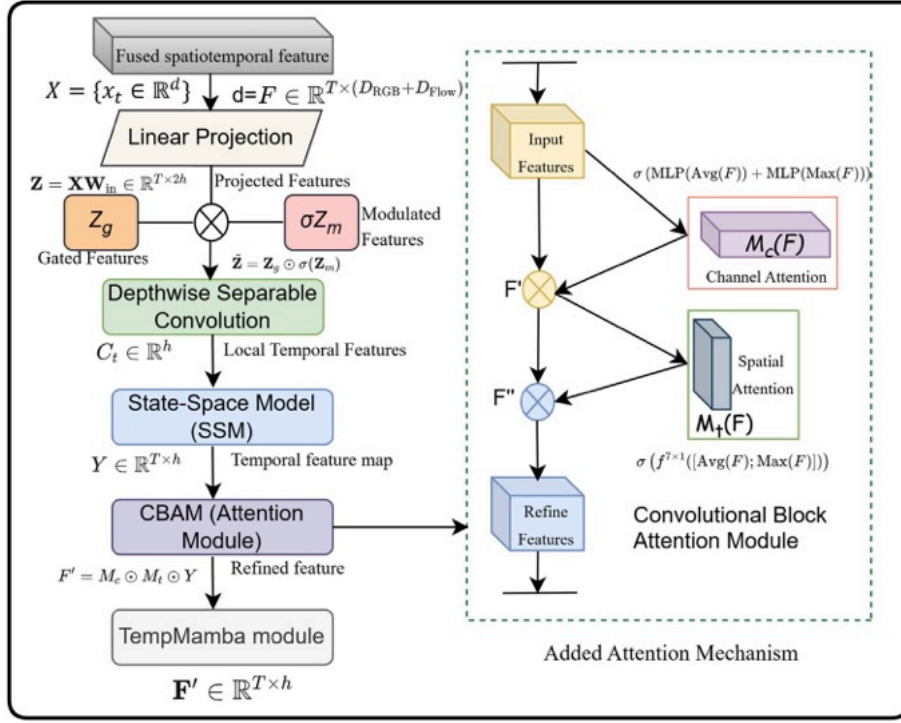


Fig. 2. Structure of the TempMamba temporal modeling unit.

to adaptively emphasize salient spatiotemporal information, ensuring a robust representation for anomaly detection.

The TempMamba module is explicitly designed to model temporal dependencies within the fused spatiotemporal feature sequence $X = \{x_t \in \mathbb{R}^d | t = 1, \dots, T\}$, where $d = D_{\text{RGB}} + D_{\text{Flow}}$. The inclusion of the Mamba block is essential for efficiently capturing both local and long-range temporal dependencies, providing a more expressive temporal representation compared to standard sequential modeling approaches. It utilizes the Mamba block for efficient and expressive temporal representation. This sequence is produced by the AugFuseNet feature fusion module, which integrates appearance and motion cues into a unified embedding. The overall architecture of the TempMamba module, responsible for modeling temporal dependencies within the fused spatiotemporal features, is illustrated in Fig. 2. The input X is first projected via a learnable linear transformation

$$Z = XW_{\text{in}}, \quad W_{\text{in}} \in \mathbb{R}^{d \times 2h}, \quad (3)$$

which is then split into two components $Z = [Z_g, Z_m]$, each in $\mathbb{R}^{T \times h}$. A gating mechanism employing the Sigmoid Linear Unit (SiLU) activation $\sigma(\cdot)$ modulates the features as $\tilde{Z} = Z_g \odot \sigma(Z_m)$, where $\odot$ denotes element-wise multiplication. While the Rectified Linear Unit (ReLU) and its variant, leaky ReLU, are widely adopted to address gradient and sparsity issues [17,18], we utilize the Sigmoid Linear Unit (SiLU) in this work. SiLU offers improved memory efficiency, stronger performance relative to ReLU, and has been shown to be effective in recent studies [16].

Depthwise separable convolutions Conv_d with kernel size k are applied to $\tilde{Z}$ to extract localized temporal patterns: $C_t = \sum_i w_i \odot \tilde{Z}_{t-i}$, where w_i are learnable convolutional kernels. To capture long-range temporal dependencies, a discrete-time linear state-space model evolves a latent state $h_t \in \mathbb{R}^m$ via the recursion $h_t = \bar{A}h_{t-1} + \bar{B}s_t$, where $\bar{A} \in \mathbb{R}^{m \times m}$ is a diagonal state transition matrix, $\bar{B} \in \mathbb{R}^m$ modulates the input $s_t = W_s C_t$, with $W_s \in \mathbb{R}^{h \times m}$. The output at time t is computed as:

$$y_t = (Ch_t) \odot c_t + D \odot c_t, \quad (4)$$

where $C, D \in \mathbb{R}^h$ are learnable parameters and c_t denotes the convolved features, followed by SiLU activation and dropout regularization to enhance robustness. To improve adaptive temporal feature representation, a CBAM is incorporated in the final layer, sequentially applying channel and temporal attention maps $M_c(\cdot)$ and $M_s(\cdot)$ to the state-space model output $Y = [y_1, \dots, y_T]$, yielding

$$F' = M_s(Y) \odot M_c(Y) \odot Y, \quad (5)$$

which enables the network to selectively focus on salient features and suppress irrelevant noise, significantly boosting anomaly detection performance. While TempMamba effectively models both local and global temporal dynamics essential for distinguishing anomalies, future work could explore multi-scale convolutional kernels to capture temporal dependencies at varying resolutions and adaptive learning of state-space parameters for enhanced flexibility across diverse anomaly patterns. Overall, the TempMamba module, augmented with CBAM, substantially improves temporal reasoning capacity and yields superior anomaly detection accuracy in complex surveillance scenarios.

3.3. MemDualNet memory-based representation learning

The MemDualNet module is proposed to enhance temporal context representation in weakly supervised video anomaly detection by leveraging a dual-memory mechanism. This design captures both short-term and long-term dependencies from the temporally enriched feature embeddings $f' \in \mathbb{R}^d$ produced by the TempMamba module. Formally, let the primary memory bank $M_p \in \mathbb{R}^{N \times d}$ and the secondary memory bank $M_s \in \mathbb{R}^{N \times d}$ be two learnable matrices, initialized randomly and updated via backpropagation. Given a sequence of temporal feature vectors $F' = \{f'_1, f'_2, \dots, f'_T\}$, the dual-memory network computes similarity scores using the Euclidean distance as follows:

$$\text{sim}_p(f') = \min_{i \in \{1, \dots, N\}} \|f' - m_i^p\|^2, \quad \text{sim}_s(f') = \min_{i \in \{1, \dots, N\}} \|f' - m_i^s\|^2, \quad (6)$$

where m_i^p and m_i^s represent the i -th memory vectors from M_p and M_s , respectively. The final output is computed as the minimum similarity across both memory banks, denoted as the Dual-Memory Deviation

Score: $\text{DDM}(f') = \min(\text{sim}_p(f'), \text{sim}_s(f'))$, which effectively quantifies the deviation of the current representation from previously learned normal patterns. This dual-memory structure allows the model to disentangle rapidly changing short-term dynamics (modeled by M_p) from stable long-term context (captured in M_s), thereby improving the detection of both transient and persistent anomalies. The memory-refined feature embeddings are then passed through a fully connected projection layer $P_{fc} : \mathbb{R}^d \rightarrow \mathbb{R}^{d'}$, yielding low-dimensional embeddings $z_i = P_{fc}(f')$ suitable for downstream tasks. Integrating MemDualNet after TempMamba substantially enhances the temporal reasoning capacity of the framework. The resulting output $\text{DDM}(f')$ provides a compact and expressive measure of temporal deviation, making it a valuable indicator for anomaly characterization in complex surveillance environments.

To prevent memory degradation or overfitting, primary memory M_p adapts rapidly to recent inputs, capturing short-term temporal variations, while secondary memory M_s stores stable long-term prototypes. Memory slots are updated using a controlled gradient update $m_i \leftarrow (1 - \lambda)m_i + \lambda \nabla m_i$ with a small learning rate. If a new embedding is significantly dissimilar from all current slots, the least-accessed slot is replaced following a frequency-aware FIFO-LRU policy. This design allows primary memory to quickly reflect recent dynamics while secondary memory maintains consistent prototypical patterns, improving interpretability and robustness of the temporal representations.

3.4. ContTrain++ continual learning framework

To address challenges posed by evolving anomaly patterns and distributional shifts in real-world surveillance streams, we introduce ContTrain++, a continual learning framework designed to maintain performance over time while minimizing catastrophic forgetting. ContTrain++ improves both temporal and semantic discriminability by combining multiple learning objectives with a memory replay mechanism.

At the heart of the framework is a replay buffer B , which stores a curated set of past samples. These buffered instances are periodically revisited during training to help the model retain prior knowledge while adapting to new data in a balanced way. During each training step, a replay batch B_r is sampled from the buffer B in addition to the current batch B_c from dataset D . To ensure that the replay batch is informative and representative, samples are selected according to $B_r = \text{Select}(B; \text{top-}k \ s_i, \text{class-balance}, \text{temporal diversity})$, where s_i denotes the anomaly score of sample $i \in B$, and the selection enforces inclusion of samples with high s_i , balanced proportions of normal and anomalous instances, and coverage of diverse temporal segments. This strategy enables the model to retain important historical knowledge while adapting to new data, mitigating catastrophic forgetting and improving robustness in continual learning. ContTrain++ then optimizes a multi-objective loss function composed of several components. Contrastive learning is incorporated to enhance representation discriminability by aligning semantically similar samples and separating dissimilar ones in the embedding space, which is critical for detecting subtle anomalies. The contrastive loss is formulated using the InfoNCE objective:

$$\mathcal{L}_{\text{CL}} = -\log \left(\frac{\exp(z \cdot z^+ / \tau)}{\sum_j \exp(z \cdot z_j / \tau)} \right) \quad (7)$$

Where z and z^+ are normalized embeddings of positive pairs, z_j represents negative samples, and τ is a temperature parameter controlling the distribution concentration. To utilize weakly labeled video data, a Multiple Instance Learning (MIL) framework is employed. It differentiates abnormal from normal video bags by focusing on the most anomalous snippets. The MIL loss encourages a margin between the highest anomaly scores in abnormal and normal bags and promotes temporal smoothness:

$$\mathcal{L}_{\text{MIL}} = \max(0, m - (s_a - s_n)) + \lambda_1 \|s_a\|_1 + \lambda_2 \|\Delta s\|_2 \quad (8)$$

Where s_a and s_n are the maximum anomaly scores in abnormal and normal bags respectively, m is the margin, and Δs captures frame-wise score differences to encourage smoothness over time.

To better handle class imbalance and focus on challenging examples, focal loss is incorporated:

$$\mathcal{L}_{\text{FL}} = -\alpha(1 - p)^\gamma \log(p) \quad (9)$$

Where p is the predicted probability for the positive (anomalous) class, α balances the class importance, and γ adjusts the down-weighting of easy samples. An anomaly score separation loss is further introduced to ensure a clear gap between the average anomaly scores of abnormal and normal video sequences:

$$\mathcal{L}_{\text{AS}} = \max(0, m - (\mu_a - \mu_n)) \quad (10)$$

Where μ_a and μ_n denote the mean anomaly scores of abnormal and normal sequences respectively. The overall loss combines all these terms:

$$\mathcal{L} = \mathcal{L}_{\text{MIL}} + \lambda_{\text{CL}} \mathcal{L}_{\text{CL}} + \lambda_{\text{FL}} \mathcal{L}_{\text{FL}} + \lambda_{\text{AS}} \mathcal{L}_{\text{AS}} + \lambda \|s\|_2 \quad (11)$$

Where the hyperparameters λ_{CL} , λ_{FL} , λ_{AS} , and λ control the contribution of each term.

By leveraging the replay buffer B , ContTrain++ samples both current and historical data during training, enabling balanced learning and mitigating forgetting. This continual adaptation supports robust weakly supervised anomaly detection in complex and dynamic environments. After training, ContTrain++ outputs an anomaly score s_i for each video snippet i , quantifying the likelihood of anomalous behavior. This score is typically computed as the maximum predicted anomaly probability among snippets in the corresponding bag:

$$s_i = \max_{i \in \text{bag}_i} \hat{y}_i \quad (12)$$

Where $\hat{y}_i$ is the predicted anomaly probability for the i -th snippet in the bag at time t . These anomaly scores enable effective temporal localization of anomalies within video streams.

3.5. Complexity analysis

In this section, we analyze the computational complexity of each core component in the proposed continual learning framework for weakly supervised video anomaly detection. To evaluate the efficiency and scalability of the proposed framework, we assess the time and space complexity of its key modules under practical deployment settings. The proposed AugFuseNet framework is designed to achieve a balance between computational efficiency and expressive capacity, enabling scalable deployment in real-world surveillance settings. Let $B = 32$ denote the batch size, $T = 32$ the temporal window, $D = 2048$ the fused feature dimension, and $M = 256$ the memory size. The feature fusion module performs modality integration via concatenation with complexity $O(B \cdot T \cdot D) \approx O(1.31 \times 10^6)$, benefiting from data augmentation to enhance generalization.

The TempMamba temporal modeling unit, incorporating Mamba blocks and CBAM attention, captures hierarchical temporal dependencies with complexity $O(B \cdot T \cdot D^2) \approx O(2.68 \times 10^9)$. Despite the quadratic term, the short sequence length T ensures computational feasibility on modern GPUs. MemDualNet computes memory-based anomaly scores over dual memory banks with complexity $O(B \cdot M \cdot D) \approx O(5.24 \times 10^6)$, supporting efficient short- and long-term contextual retrieval. The ContTrain++ continual learning module utilizes a replay buffer of size $R = 1000$, adding minimal overhead of $O(B \cdot D) \approx O(4.1 \times 10^4)$ per step. Its composite loss function, comprising contrastive, MIL, focal, and anomaly score separation losses, is optimized with batch-level gradient updates. Overall space complexity is bounded by $O(D^2 + B \cdot T \cdot D) \approx O(5.5 \times 10^6)$ floats (excluding buffer), scaling linearly with replay memory. The design ensures real-time inference with competitive performance, supporting robust deployment in complex surveillance environments.

4. Experiment

4.1. Datasets

To rigorously evaluate the effectiveness and generalizability of our proposed method across diverse surveillance scenarios, we employ three widely adopted benchmarks: UCF-Crime [48], XD-Violence [60] and ShanghaiTech [27]. These datasets provide complementary evaluation paradigms, with UCF-Crime emphasizing long-duration events across diverse anomaly categories, while ShanghaiTech focuses on scene variations and fine-grained temporal anomalies. Additionally, XD-Violence enables targeted assessment of violent behavior modeling across six major subclasses, including physical aggression and crowd violence, further validating the framework's capacity to discriminate and localize sparse but critical anomalous events. Together, this multi-dataset evaluation supports a comprehensive analysis of spatiotemporal representation learning and continual learning stability under weak supervision.

The UCF-Crime dataset represents one of the most extensive real-world surveillance benchmarks for anomaly detection, consisting of 1900 untrimmed videos evenly distributed between 950 anomalous and 950 normal samples. It encompasses 13 distinct anomaly classes including violent activities (Assault, Abuse), property crimes (Burglary, Vandalism), and hazardous incidents (Explosion). Captured in unconstrained outdoor environments, the dataset presents significant challenges due to high intra-class variability in abnormal events, complex temporal dynamics in extended video sequences, and diverse motion patterns under varying illumination conditions.

In contrast, the ShanghaiTech Campus dataset comprises 437 surveillance videos (130 anomalous, 307 normal) collected from 13 distinct camera perspectives within an academic environment. This dataset features short-term, fine-grained anomalies such as unusual pedestrian behaviors (running, loitering) and suspicious actions (object throwing). The relatively static backgrounds, combined with sparse temporal occurrences of abnormal events, make this dataset particularly valuable for evaluating model generalization under weak supervision and adaptability to different environmental contexts.

As an additional dataset, XD-Violence is a large-scale weakly supervised benchmark containing 4754 untrimmed videos (217 hours), with 2405 violent and 2349 normal samples. The dataset spans diverse environments including CCTV footage and online video sources, and supports audio-visual learning for detecting sparse, high-risk anomalies such as fights, riots, accidents, explosions, and weapon-related events. It provides a challenging testbed for evaluating long-term adaptation and cross-modal anomaly discrimination under video-level supervision.

The complementary nature of these datasets ensures thorough evaluation across multiple dimensions. UCF-Crime evaluates robustness on diverse, long-duration anomalies in complex outdoor scenes, ShanghaiTech measures sensitivity to subtle, short-term deviations in controlled environments, and XD-Violence further tests cross-modal anomaly discrimination for sparse, high-risk incidents. Collectively, they form a comprehensive testbed for assessing anomaly separability, temporal modeling capacity, and long-term adaptability across evolving surveillance scenarios.

4.2. Evaluation metrics

To quantitatively evaluate anomaly detection performance under weak supervision, we adopt the frame-level Area Under the Receiver Operating Characteristic Curve (AUC) as the primary metric. The ROC curves illustrate the trade-off between the true positive rate and false positive rate across different decision thresholds, providing a comprehensive view of the model's discrimination capability. We report both the overall AUC and the mean AUC calculated across all test videos, offering an aggregate measure of detection performance over diverse scenes and anomaly classes. Additionally, subclass-level AUCs are computed to assess the method's effectiveness across different anomaly

categories. As a complementary ranking-based metric, we further report Average Precision (AP), which summarizes the area under the precision-recall curve and better reflects detection reliability on highly imbalanced anomaly distributions. These evaluation metrics collectively ensure a rigorous and balanced assessment of frame-level classification accuracy and generalization performance, consistent with standard practices in video anomaly detection.

Table 2

AUC comparison on the UCF-Crime dataset.

Supervision	Method	AUC (%)
Weakly Supervised	Sultani et al. (2018) [48]	75.41
	Park et al. (2020) [34]	84.70
	Zaheer et al. (2020) [67]	83.03
	Tian et al. (2021) [52]	84.03
	Feng et al. (2021) [15]	82.30
	Sapkota and Yu (2022) [43]	83.39
	Li et al. (2022) [25]	85.30
	Chen et al. (2023) [8]	86.98
	Chen et al. (2024) [7]	86.83
	Zhang et al. (2024) [70]	80.60
	Majhi et al. (2024) [31]	84.33
	Jain et al. (2025) [23]	84.73
	Yang et al. (2025) [65]	83.10
	Wang et al. (2025) [57]	84.70
	COMMAND	87.23

Table 3

AUC comparison on the ShanghaiTech dataset.

Supervision	Method	AUC (%)
Weakly supervised	Wan et al. (2020) [53]	91.24
	Feng et al. (2021) [15]	94.83
	Tian et al. (2021) [52]	91.51
	Mu et al. (2022) [33]	92.30
	Zhang et al. (2024) [70]	95.70
	Su et al. (2024) [47]	96.88
	Majhi et al. (2024) [31]	93.72
	Su et al. (2025) [46]	97.10
	Yang et al. (2025) [65]	96.50
	COMMAND	97.37

Table 4

AP comparison on the XD-Violence.

Supervision	Method	AP (%)
Weakly supervised	Sultani et al. (2018) [48]	73.20
	Tian et al. (2021) [52]	77.81
	Li et al. (2022) [25]	78.28
	Wu et al. (2022) [59]	78.64
	Chen et al. (2023) [8]	79.19
	Zhang et al. (2023) [69]	78.74
	Park et al. (2023) [35]	78.51
	He et al. (2024) [20]	80.07
	Zanell et al. (2024) [68]	78.51
	Yang et al. (2025) [64]	81.08
	Zhong et al. (2025) [73]	80.02
	Paulraj et al. (2025) [36]	78.06
	COMMAND	83.14

Table 5

Cross-dataset(UCF-Crime and ShanghaiTech) AUC evaluation of our method.

Train Dataset	Test Dataset	AUC (%)
UCF-Crime	UCF-Crime	87.23
ShanghaiTech	ShanghaiTech	97.37
UCF-Crime	ShanghaiTech	96.51
ShanghaiTech	UCF-Crime	85.82

Table 6
Subclass-level AUC scores (%) of different methods on the UCF-Crime dataset.

Method	Abus	Arre	Arso	Assa	Burg	Expl	Figh	Road	Robb	Shoo	Shop	Stea	Vand	Avg
CL-WSVAD	99.9	98.1	97.5	95.3	92.7	88.7	88.6	87.6	86.7	84.5	83.2	83.2	83.1	89.9
BiLSTM	98.1	93.4	95.6	95.9	84.1	85.3	93.2	92.6	95.2	93.5	89.5	90.5	88.7	92.0
Mamba	99.1	93.2	94.9	97.8	97.2	88.6	96.2	97.1	96.8	95.3	91.2	91.5	92.3	94.7
COMMAND	99.3	97.4	94.5	99.6	96.0	90.2	99.0	97.3	98.2	94.8	95.1	95.9	96.5	96.4

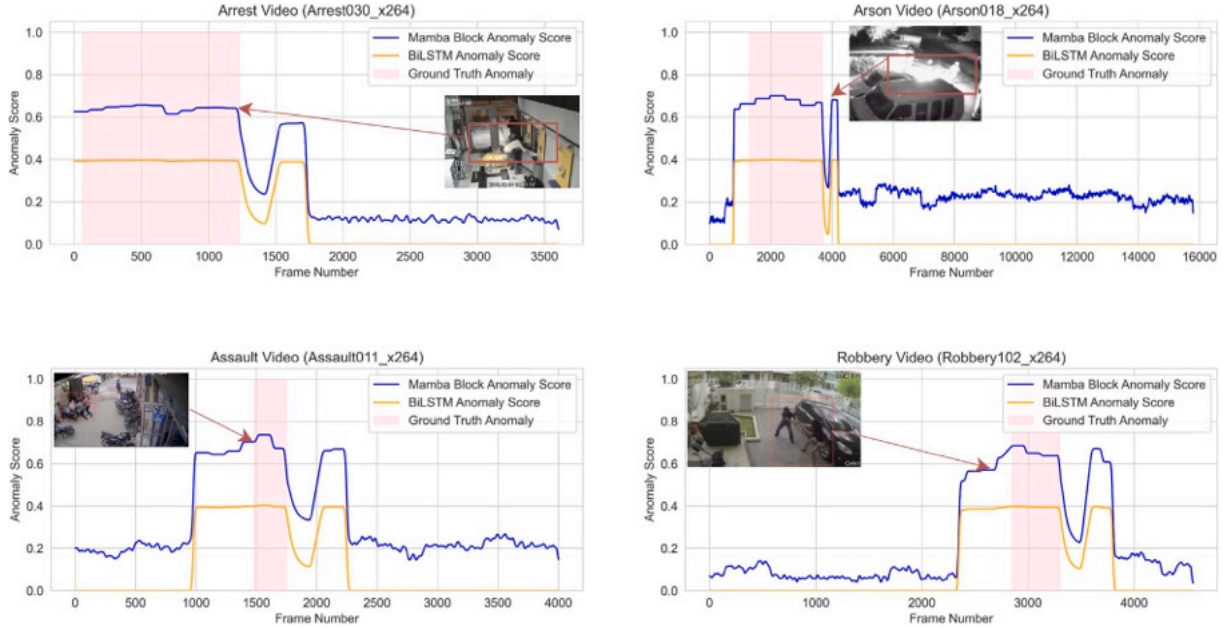


Fig. 3. Mamba generates sharp, selective anomaly score spikes, compared to smoother BiLSTM responses.

4.3. Implementation details

The proposed framework is implemented using PyTorch and trained on a workstation equipped with an Intel® Core™ i5 12600KF CPU at 3.70 GHz, 32 GB of RAM, and an NVIDIA RTX 3090 GPU, running Windows 11. We utilize pre-extracted features obtained from an I3D network pre-trained on the Kinetics dataset, leveraging both RGB and optical flow modalities. Optical flow is computed using the Farneback method prior to feature extraction. Data augmentation including random horizontal flips, cropping, and spatial rotations is applied at the feature level to improve generalization. The model is trained for 100 epochs using the Adam optimizer with an initial learning rate of 1×10^{-4} , reduced by a factor of 0.1 every 10 epochs, and a batch size of 32. The TempMamba module employs a temporal kernel size of $k = 5$, while the MemDualNet module maintains dual memory banks of size $N = 256$. For continual learning, ContTrain++ uses a replay buffer containing 1000 samples and a composite loss function with weights $\lambda_{CL} = 1.0$, $\lambda_{MIL} = 1.0$, $\lambda_{FL} = 0.5$, and $\lambda_{AS} = 0.3$. All hyperparameters are selected via grid search on a held-out validation set. While the composite loss function uses the weights above, we empirically observed that small variations in λ_{CL} (e.g., ± 0.2) had minimal impact on overall performance, resulting in only about a 1% decrease in AUC. This indicates that the model is stable with respect to the continual learning weight. A more extensive hyperparameter sensitivity analysis is left for future work.

4.4. Comparative performance analysis

To rigorously evaluate the effectiveness and generalizability of the proposed continual learning framework for weakly supervised video anomaly detection, we conduct comprehensive comparative

experiments against a diverse set of state-of-the-art methods. These evaluations cover weakly supervised paradigms across the UCF-Crime, ShanghaiTech, and XD-Violence datasets, with results summarized in Tables 2–4, respectively. Table 2 demonstrates that the proposed method surpasses existing approaches on the UCF-Crime benchmark, outperforming recent competitive baselines such as [7,8,25], and [57]. This highlights the framework’s robustness in managing large-scale surveillance data characterized by complex and evolving anomaly patterns under weak supervision.

Similarly, as shown in Table 3, the method achieves state-of-the-art performance on the ShanghaiTech dataset, outperforming notable recent techniques including [31,47], and [65]. These results underscore the framework’s strong generalization capabilities across different data domains and anomaly types. Collectively, the experimental findings substantiate the superiority and adaptability of the proposed approach relative to contemporary methods in video anomaly detection.

In addition, the proposed method is evaluated on the XD-Violence dataset, which poses a more challenging benchmark due to its diverse violent events, multi-modal inputs, and long-duration untrimmed videos. As shown in Table 4, our approach achieves state-of-the-art performance in terms of average precision (AP), surpassing recent weakly supervised methods such as [37,64]. These results further highlight the robustness and generalization capability of the framework in complex real-world surveillance scenarios.

Despite our framework’s superior performance on both the UCF-Crime and ShanghaiTech datasets, a limitation arises under cross-dataset evaluation. When transferred directly to an unseen dataset without fine-tuning, the model experiences a slight drop in performance, as reported in Table 5. This indicates that while our learned representations are robust and maintain high accuracy, they are not fully

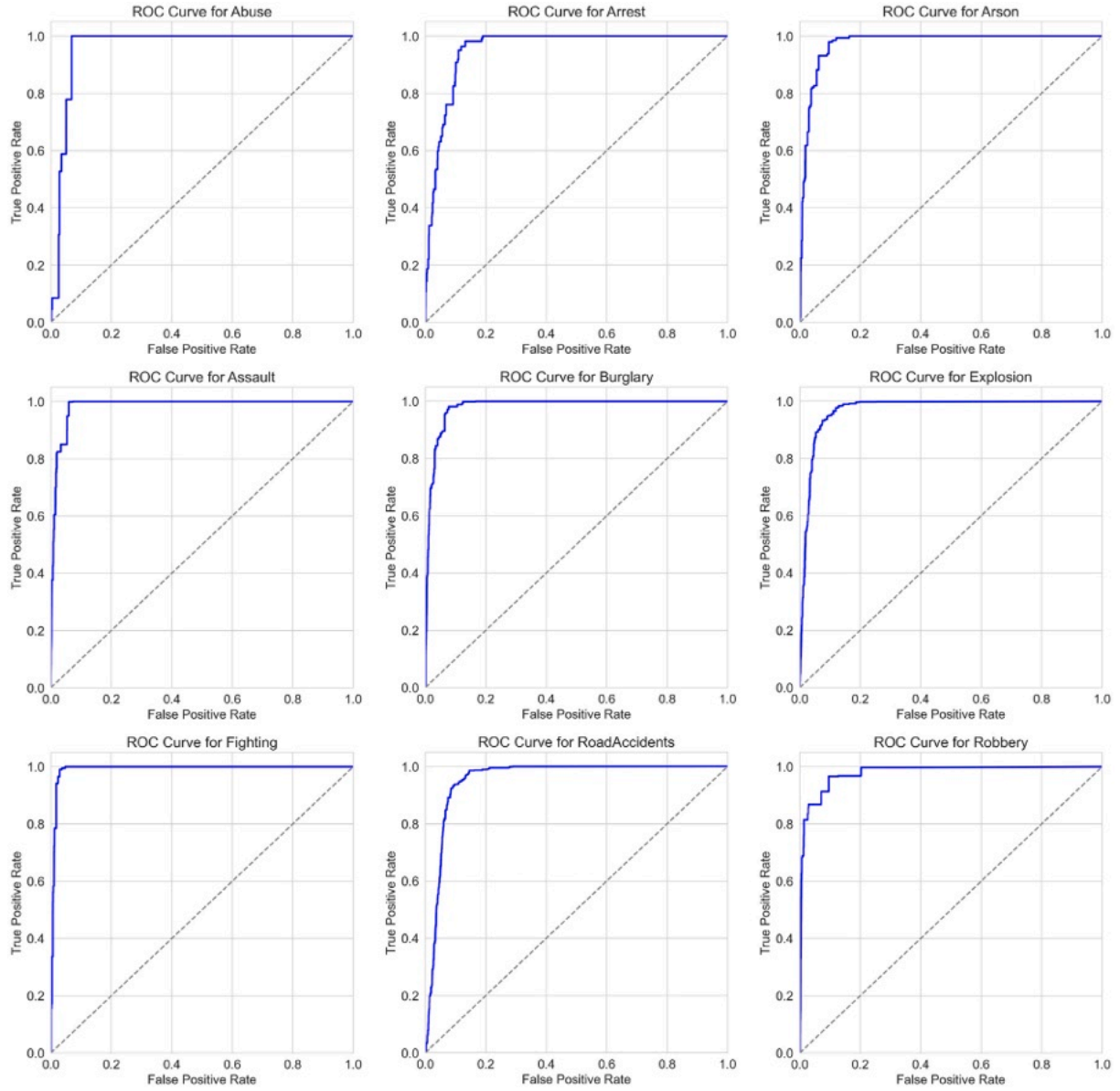


Fig. 4. Subclass-wise ROC curves for AUC evaluation on UCF-Crime.

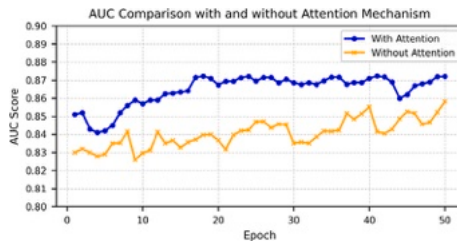


Fig. 5. Impact of attention mechanism on UCF-Crime.

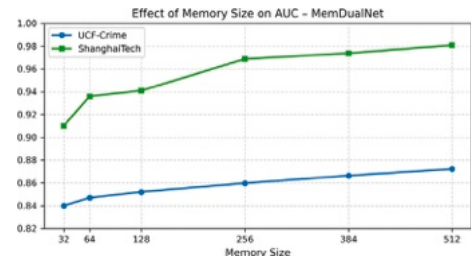


Fig. 6. Impact of memory size in MemDualNet-based anomaly detection.

invariant to domain shifts. Importantly, the within-dataset results remain state-of-the-art, demonstrating strong discriminative capability. We also note that the framework relies on a high-performance GPU for real-time inference, which may challenge deployment on resource-constrained devices. Future work will investigate domain adaptation and model distillation techniques to further enhance generalizability across diverse environments.

4.5. Performance evaluation across anomaly subclasses

To rigorously assess the effectiveness of our proposed method at the subclass level, we conduct detailed experiments on the 13 predefined categories of the UCF-Crime dataset. Our investigation begins with a BiLSTM-based model that forms the foundation of our approach,

Table 7

Subclass-level AUC scores (%) for different temporal window sizes on the UCF-Crime dataset.

Size	Abus	Arre	Arso	Assa	Burg	Expl	Figh	Road	Robb	Shoo	Shop	Stea	Vand	Avg
16	87.1	85.1	91.0	96.5	82.0	89.3	93.2	83.5	83.0	73.3	84.7	92.3	92.1	87.1
20	92.4	94.9	93.5	98.4	93.9	70.4	89.6	86.8	98.5	92.9	90.5	94.2	90.7	91.3
24	97.2	92.7	93.6	98.1	95.3	86.2	95.2	96.5	97.9	93.7	92.2	93.6	92.4	94.4
32	99.3	97.4	94.5	99.6	96.0	90.2	99.0	97.3	98.2	94.8	95.1	95.9	96.5	96.4
40	99.1	97.9	97.2	99.8	97.4	91.5	98.7	98.8	98.9	96.7	97.1	97.5	95.8	97.4
48	99.4	98.5	96.2	99.1	98.7	96.5	97.2	98.5	99.1	98.6	97.3	98.1	96.2	97.9

Table 8

Effect of temporal window size on anomaly detection performance.

Window size	UCF-Crime	ShanghaiTech
8	80.21	87.58
16	83.68	91.03
24	85.46	94.35
32	87.23	97.37
40	87.93	98.12
48	88.42	98.17

Table 9

AUC (%) of ContTrain++ with varying replay buffer sizes, along with average memory usage (MB).

Replay Buffer Size	UCF-Crime	ShanghaiTech	Avg. Memory (MB)
No Buffer	84.12	95.36	118
Buffer (500)	86.62	96.54	144
Buffer (1000)	87.23	97.37	175

effectively capturing temporal dynamics and delivering improved consistency across anomaly types. To better model long-range temporal dependencies, we extend this architecture with a Mamba-based sequential backbone, resulting in notable gains in subclass-level detection performance. Building upon this, we integrate the Convolutional Block Attention Module (CBAM) into the Mamba architecture, enabling the model to adaptively emphasize salient spatiotemporal features while suppressing irrelevant information. As evidenced by the experimental results presented in Table 6, this attention-enhanced variant consistently achieves the highest average AUC across all categories, underscoring the benefits of combining structured temporal modeling with dynamic attention mechanisms. In comparison, while CL-WSVAD [65], BiLSTM, and Mamba achieve competitive AUC scores on certain subclasses, their performance tends to be less consistent across the full range of categories relative to the COMMAND (Mamba + CBAM) variant, which demonstrates both higher average accuracy and greater stability. Overall, our proposed TempMamba framework delivers more stable and superior detection performance across the full spectrum of anomalous events, establishing its effectiveness for weakly supervised video anomaly detection.

4.6. Comprehensive ROC evaluation of anomaly detection performance

To rigorously evaluate the efficacy of the proposed continual learning framework, we perform a detailed analysis on the UCF-Crime dataset encompassing 13 distinct anomaly categories. Fig. 4 illustrates the Receiver Operating Characteristic (ROC) curves for each category, showcasing the model's capability to distinguish between normal and abnormal events across diverse anomaly types. The ROC curves depict the trade-off between true positive rate and false positive rate at varying thresholds, offering a comprehensive measure of detection accuracy. The area under the ROC curve (AUC) serves as a quantitative metric of this discriminative ability, with higher values indicating superior anomaly detection performance. The consistently strong AUC values observed

across all categories underscore the robustness and generalizability of the framework in weakly supervised settings.

4.7. Temporal selectivity evaluation of the mamba backbone

Although Mamba provides efficient long-range temporal modeling, its selective state-space updates are especially valuable for anomaly detection, where abnormal events are sparse and temporally unpredictable. Unlike BiLSTM models that propagate historical information uniformly, Mamba updates its latent temporal state conditioned on input relevance, amplifying abrupt deviations from learned normal dynamics. Fig. 3 empirically validates this behavior on the UCF-Crime dataset, showing that the Mamba backbone produces sharp and localized anomaly score spikes aligned with ground-truth anomalous segments. This confirms the dynamic temporal focus enabled by Mamba's selective state evolution, which is particularly effective for detecting short-lived and irregular anomalies in long video streams.

4.8. Effect of CBAM attention module on TempMamba temporal modeling

The TempMamba module, based on the MambaBlock architecture, captures both short- and long-range temporal dependencies within the fused spatiotemporal feature sequence $X = \{x_t \in \mathbb{R}^d | t = 1, \dots, T\}$, where $d = D_{\text{RGB}} + D_{\text{Flow}}$. To enhance temporal feature representation, the Convolutional Block Attention Module (CBAM) is incorporated before the final output layer. CBAM applies channel and temporal attention maps $M_c(\cdot)$ and $M_s(\cdot)$ to the output Y , producing the refined embedding $F' = M_s(Y) \odot M_c(Y) \odot Y$. As shown in Fig. 5, integrating CBAM consistently improves the AUC performance on both UCF-Crime and ShanghaiTech datasets compared to the base TempMamba model. This improvement demonstrates CBAM's effectiveness in emphasizing salient temporal regions while suppressing irrelevant features, resulting in more accurate anomaly detection under weak supervision.

4.9. Impact of memory size in MemDualNet-based anomaly detection

To enhance temporal reasoning and reinforce the representation of normal behavioral patterns, we integrate MemDualNet, a memory-based representation learning module that maintains a bank of prototypical normal features. This mechanism enables the model to effectively recall and compare current observations with stored representations, thereby improving its capacity to detect deviations under weak supervision. We perform experiments on the UCF-Crime and ShanghaiTech datasets using varying memory sizes of 32, 64, 128, 256, 384, and 512. As illustrated in Fig. 6, increasing the memory size results in consistent improvements in detection performance, demonstrating that larger memory banks provide greater representational capacity and more reliable anomaly localization. Although the best performance is observed at a memory size of 512, this comes with increased computational overhead. Consequently, we select a memory size of 256 as a balanced choice that achieves strong detection accuracy while maintaining computational feasibility for deployment in practical surveillance scenarios.

Table 10
Continual learning performance (AUC %) on UCF-Crime.

Method	Task1 (T1/T2/T3)	Task2 (T2/T3)	Task3 (T3)	Avg Forgetting
UCAD [26]	82.3±0.5 / 76.4±0.6 / 70.0±0.7	81.7±0.5 / 74.1±0.6	80.4±0.5	10.0
GDumb [38]	80.3±0.6 / 73.6±0.7 / 68.2±0.7	78.9±0.5 / 72.1±0.6	77.4±0.6	9.4
Dark Experience Replay [4]	83.2±0.5 / 77.8±0.6 / 72.2±0.6	82.8±0.5 / 76.4±0.6	81.5±0.5	8.5
No Buffer	84.1±0.5 / 77.5±0.6 / 71.3±0.7	83.9±0.5 / 76.8±0.6	83.2±0.5	9.5
Buffer (500)	86.6±0.4 / 82.7±0.5 / 79.4±0.5	85.8±0.4 / 81.5±0.5	84.7±0.4	4.8
ContTrain++ (1000) (Ours)	87.1±0.3 / 84.3±0.4 / 81.8±0.4	86.4±0.3 / 82.9±0.4	85.6±0.3	3.2

Table 11
Inference efficiency evaluation.

Method	T.T (min/epoch)	P.G.Mem (GB)	Time (ms/samp.)			Throughput (samp./s)			Mem. (MB)			Latency (ms)	FPS
			8	16	32	8	16	32	8	16	32		
BiLSTM	0.52	3.8	0.75	0.65	0.48	1333	1538	2083	115	120	130	5.2	192
Mamba	0.46	4.6	0.60	0.31	0.20	1600	3226	5000	125	140	170	4.8	208
RTFM[52]	1.20	7.1	1.20	0.95	0.70	833	1053	1428	210	235	280	5.5	182
COMMAND (Ours)	0.34	5.25	0.58	0.29	0.18	1724	3420	5556	128	144	175	4.3	232

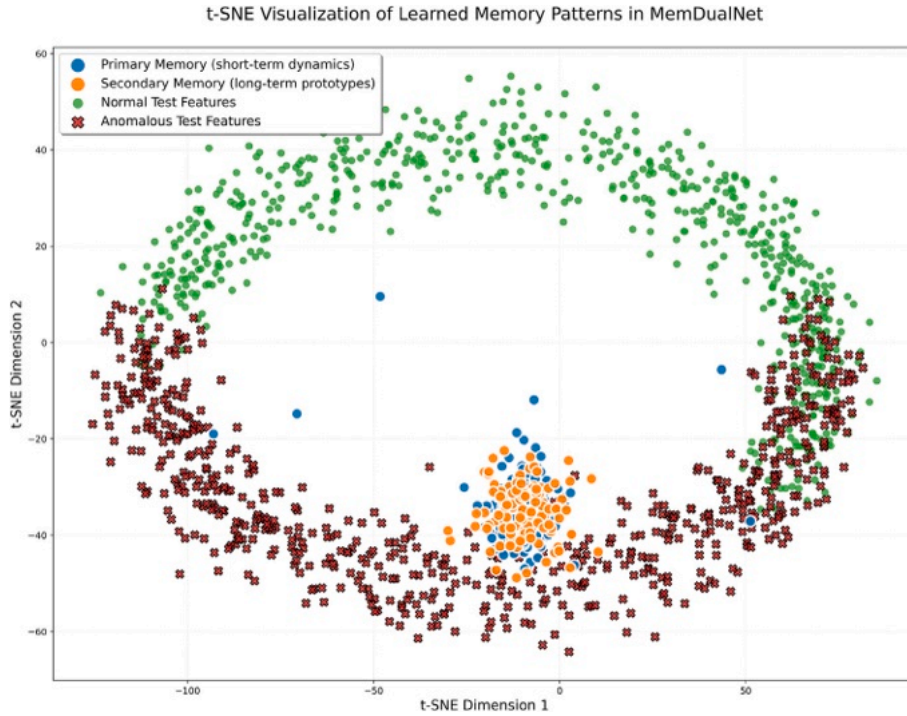


Fig. 7. t-SNE visualization of MemDualNet memory banks, illustrating fast adaptation to recent patterns and stable long-term prototypes for improved interpretability in anomaly detection.

4.10. Impact of temporal window size on detection performance

To evaluate the temporal modeling capacity of the proposed TempMamba module, we systematically examine the effect of varying temporal window sizes (8, 16, 24, 32, 40, and 48 frames) on anomaly detection performance. The class-wise AUC results on the UCF-Crime dataset are reported in Table 7, covering 13 anomaly subclasses. The results show a clear performance improvement with larger temporal windows, as extended sequences allow TempMamba to capture richer temporal dependencies. This is especially advantageous for complex scenarios such as Assault, Burglary, and Robbery, where longer context helps distinguish subtle abnormal patterns. These findings validate the effectiveness of TempMamba in learning discriminative spatiotemporal representations under weak supervision.

Additionally, Table 8 shows the overall AUC scores for both the UCF-Crime and ShanghaiTech datasets across different window sizes.

While the best detection performance is observed at 48 frames, using larger temporal windows introduces greater computational cost and inference latency. Therefore, to achieve a practical trade-off between accuracy and efficiency, we choose a window size of 32 frames for the remaining experiments. This configuration offers a favorable balance, delivering strong detection results while maintaining runtime feasibility for real-world deployments.

4.11. Impact of replay buffer size on ContTrain++ performance

Continual learning systems are inherently susceptible to catastrophic forgetting, particularly in dynamic video anomaly detection scenarios. To mitigate this, the proposed ContTrain++ framework integrates a replay buffer mechanism that selectively retains representative samples from previous data distributions. This design choice facilitates

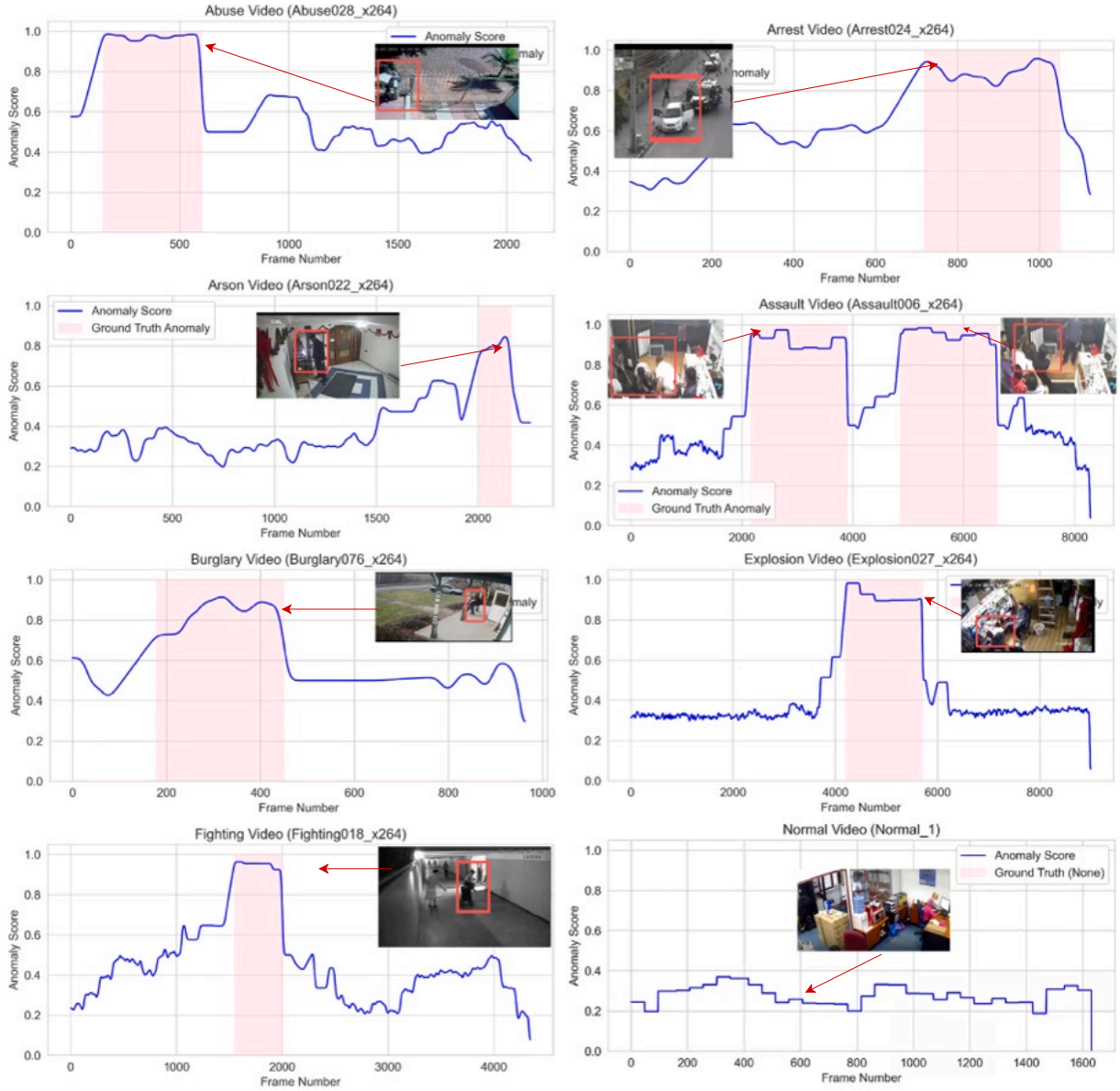


Fig. 8. Qualitative analysis on UCF-Crime demonstrating anomaly localization.

more stable model updates and improves long-term knowledge retention during incremental learning.

To assess the effectiveness of varying replay buffer sizes, we perform a controlled evaluation on both the UCF-Crime and ShanghaiTech datasets. As shown in Table 9, we compare three configurations: no replay buffer, a buffer of 500 samples, and a buffer of 1000 samples. The results clearly demonstrate that incorporating a replay buffer yields consistent improvements in AUC performance. In particular, the configuration with 1000 samples achieves the highest scores, indicating that a larger memory capacity enhances the model's ability to preserve learned representations. Based on this analysis, we adopt a buffer size of 1000 as a balanced choice between performance and computational overhead.

4.12. Evaluation of continual learning and catastrophic forgetting

To evaluate the Continual Learning (CL) capabilities of ContTrain++ and its resistance to catastrophic forgetting, we conducted a task-incremental experiment on the UCF-Crime dataset. The 13 anomaly

classes were divided into three sequential tasks based on semantic similarity: Task 1 (Abuse, Arrest, Arson, Assault), Task 2 (Burglary, Explosion, Fighting, Road), and Task 3 (Robbery, Shooting, Shoplifting, Stealing, Vandalism). Tasks are defined based on anomaly types and are presented sequentially to the model, simulating a continual learning scenario. The model was trained sequentially on each task, and its performance was measured on all tasks after each training phase to quantify knowledge retention and forgetting. Table 10 presents the results, comparing ContTrain++ to three continual learning baselines: UCAD [26], GDumb [38], and Dark Experience Replay [4], as well as no-replay and buffer-based variants. The results, averaged across three independent trials, show that ContTrain++ with a replay buffer of 1000 samples achieves the highest AUC and lowest average forgetting, demonstrating its effectiveness in maintaining robust performance across sequential tasks in weakly supervised video anomaly detection. In the task-incremental setting, new training streams are assumed to include weak video-level labels, enabling continual adaptation without requiring frame-level annotations. This assumption reflects realistic

long-term surveillance operations, where coarse labels are feasible and annotation costs must remain low.

4.13. Inference efficiency evaluation

To assess the practical efficiency of our framework, we measure both training and inference performance, comparing a standard BiLSTM, a base Mamba model, the Transformer-based RTFM [52], and our final COMMAND framework. Experiments are conducted at batch sizes 8, 16, and 32, recording time per sample, throughput, memory usage, training time per epoch, single-sample inference latency, maximum sustainable frame rate (FPS) and peak GPU memory. Results are presented in Table 11. COMMAND consistently achieves lower inference latency and higher throughput compared to BiLSTM and Mamba, while maintaining moderate memory consumption. Training time and GPU usage remain practical for all evaluated batch sizes. These results demonstrate that COMMAND provides an efficient trade-off between detection accuracy and computational requirements, supporting deployment in real-world and resource-constrained environments such as edge cameras.

4.14. MemDualNet memory pattern analysis

In this section, we analyze MemDualNet, which employs two learnable memory banks to capture complementary normal patterns from TempMamba features. The primary memory adapts quickly to recent inputs, encoding short-term dynamics, while the secondary memory evolves conservatively to preserve stable long-term prototypes. Both are updated via backpropagation with differentiated learning rates to balance adaptability and stability. Each memory slot is refined gradually to prevent degradation or overfitting. For a feature vector, the dual-memory deviation score is computed as the minimum distance to either memory bank, providing a compact anomaly indicator. Fig. 7 visualizes the learned memory distributions using t-SNE on the UCF-Crime dataset, illustrating how the primary memory captures adaptive short-term patterns, while the secondary memory maintains stable long-term prototypes (Fig. 8).

4.15. Qualitative analysis

A comprehensive qualitative analysis was conducted to visually assess COMMAND's performance on the challenging UCF-Crime dataset, as shown in Fig. 7. As demonstrated by our visual results, our model successfully detects a variety of anomalous events in complex, real-world scenarios, such as abuse and arrests. It accurately identifies and localizes the anomalous events by producing a high anomaly score and a corresponding heatmap. Our model exhibits remarkable stability and robustness on normal videos, with consistently low anomaly scores, which is a critical factor for a reliable real-world video surveillance system. This qualitative analysis confirms that COMMAND not only achieves superior quantitative performance but also demonstrates a robust capacity for authentic anomaly localization in complex, unconstrained environments.

5. Conclusion

This work introduces a novel continual learning framework tailored for weakly supervised video anomaly detection, explicitly designed to address the challenges posed by unseen and evolving anomalies in real-world surveillance settings. By synergistically combining multimodal feature fusion through AugFuseNet, sophisticated temporal modeling via TempMamba, a dual-memory representation in MemDualNet, and an adaptive continual learning strategy in ContTrain++, the proposed approach effectively enhances anomaly detection performance under limited supervision. Comprehensive evaluations on benchmark datasets substantiate the framework's superior accuracy, temporal reasoning, and generalization capabilities compared to state-of-the-art methods. The

unified architecture not only mitigates catastrophic forgetting but also facilitates seamless adaptation to new anomaly distributions, rendering it highly suitable for practical, large-scale surveillance deployments. Future research directions include refining the continual learning mechanisms and extending the framework to incorporate additional contextual cues, with the aim of further improving robustness and detection efficacy in complex, dynamic environments. In addition, integrating large language model-guided anomaly reasoning could enhance interpretability and adaptability, while exploring other domains such as medical or industrial anomaly detection may further demonstrate the framework's generalization capabilities.

CRedit authorship contribution statement

Yan Liu: Writing – original draft. **Kaiju Li:** Resources, Conceptualization. **Md Sabuj Khan:** Methodology, Conceptualization. **Jian Lang:** Project administration, Methodology. **Rongpei Hong:** Methodology, Data curation. **Kunpeng Zhang:** Supervision. **Fan Zhou:** Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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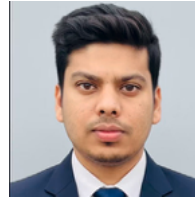
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